

Unit 1, Lesson 3: Multiplying Fractions

Benchmark

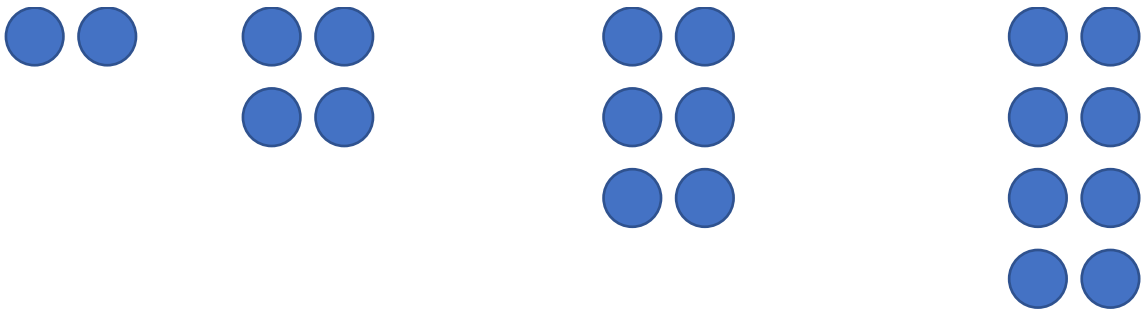
MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction or multiplication of fractions, including mixed numbers and fractions greater than 1.

Learning Target

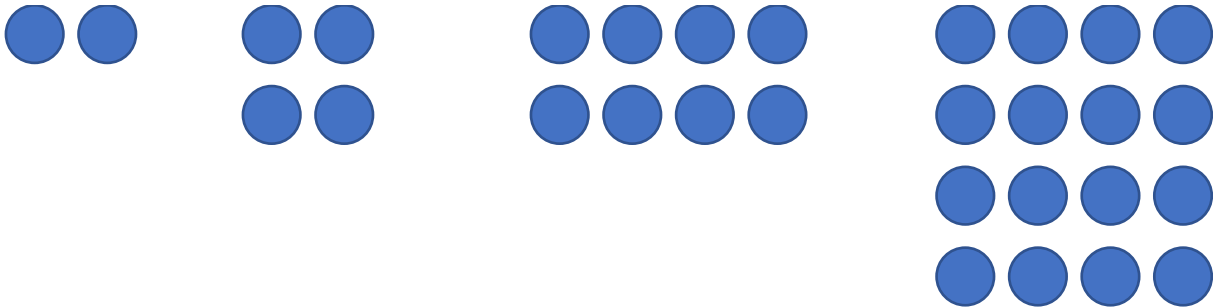
- ☐ I can solve real-world problems involving multiplication of fractions.

1.3.1 Warm-Up: Patterns

Pattern 1



Term 1 Term 2 Term 3 Term 4



Pattern 2

- 1. What do you notice about the terms in Pattern 1?
- 2. What do you notice about the terms in Pattern 2?
- 3. Write an expression to represent each pattern. Use x for the term number.

Pattern 1	Pattern 2

1.3.2 Guided Instruction: Multiplying Fractions

Alejandro took a trip to Key West, Florida. While there, he ate 2 slices of Key Lime Pie. He enjoyed the pie so much that he decided to make some pies for his family. The ingredients for one Key Lime Pie are shown.

Key West’s Famous Key Lime Pie		
1 graham cracker crust	$\frac{1}{2}$ cup sour cream	1 tablespoon grated lime zest
3 cups sweetened condensed milk	$\frac{3}{4}$ cup key lime juice	

1. Alejandro wanted to make 5 Key Lime Pies to share with his family.
- a. Write an expression that can be used to determine how much sour cream Alejandro should buy.

b. Multiplication can be thought of as

repeated addition.

repeated subtraction.

c. Rewrite your expression using the operation selected.

d. Draw a model to represent the expression.

2. Alejandro's grandma, Camila, sliced her pie to share with her friends. She cut the pie into 8 slices.

a. Draw a model of the pie cut into 8 slices below.

- b. Camila gave her friend Liliana $\frac{1}{4}$ of the pie. Use your model to show how many slices Liliana received.

- c. Complete a multiplication expression whose value is the amount of pie that Liliana received.

$$\boxed{} \times \boxed{}$$

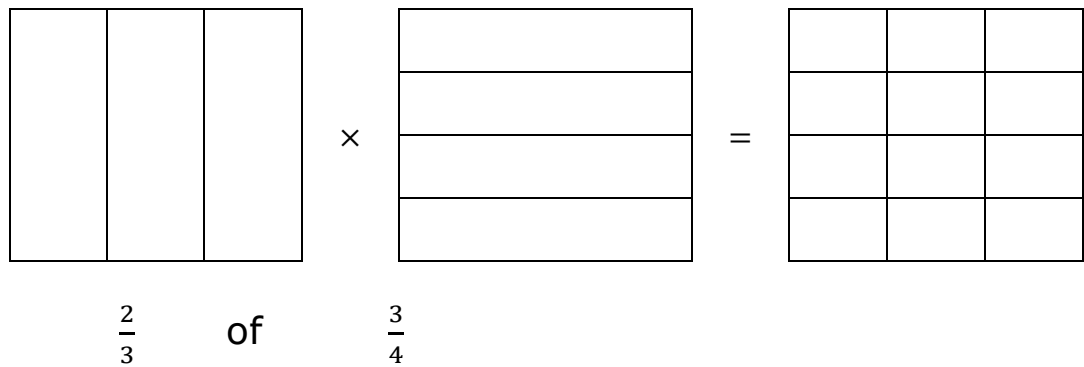
- d. How many slices of the pie are left?

- e. Complete a multiplication expression whose value represents the amount of pie that remains after Liliana received her slices.

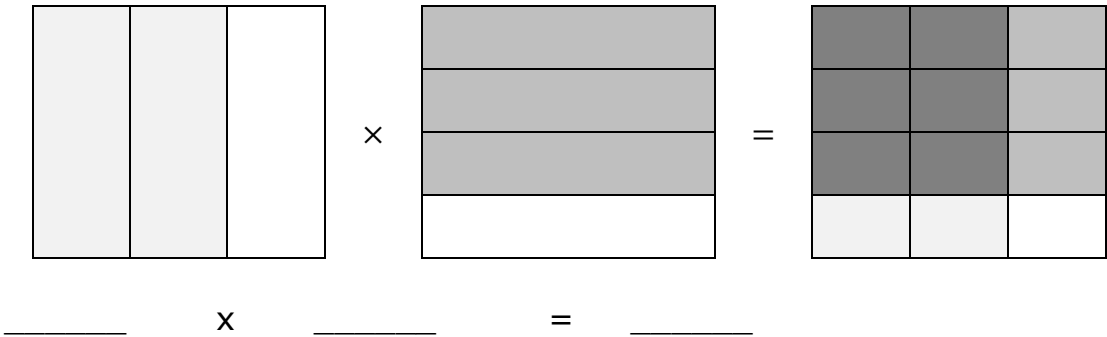
$$\boxed{} \times \boxed{}$$

3. Alejandro wanted to make a sixth pie but realized that he only had $\frac{2}{3}$ of the amount of key lime juice that he needed.

- a. Below is an incomplete model that represents the amount of key lime juice that Alejandro has. Shade in the model to represent the context.



- b. The diagram of three figures shows the complete model that represents the amount of key lime juice that Alejandro has. When the first two figures in the diagram are placed on top of each other, the result would be the third figure. In the third figure, the denominator represents the total parts. The numerator is represented by the number of parts where the shading overlaps. Fill in the fractions to complete the equation below the model.



- c. How much key lime juice did Alejandro have?

4. When multiplying fractions, the numerator of the first factor is multiplied by the

☐ numerator

☐ denominator

 of the second factor. The denominator of the first factor is multiplied by the

☐ numerator

☐ denominator

 of the

second factor. Simply put, when multiplying fractions,

multiply
☐ straight across.
☐ diagonally (cross-multiply).

5. Alejandro knows he needs $7\frac{1}{2}$ cups of key lime juice to make a batch of 10 pies. He wants to make $2\frac{4}{5}$ batches for an upcoming community bake sale.

Complete the statement about how many cups of key lime juice Alejandro needs to make the pies for the bake sale.

To find the total number of cups of key lime juice

he needs, Alejandro should
☐ add
☐ multiply

☐ $2\frac{4}{5}$
☐ 10

with $7\frac{1}{2}$ because he needs
☐ $2\frac{4}{5}$
☐ 10
 batches.

6. When multiplying using mixed numbers, an area model can be used.
- a. Complete the left side of the area model with the missing factors, and then find the product for each box.

	7	$\frac{1}{2}$

- b. Add the values from each box together to find the total number of cups of key lime juice.

1.3.3 Guided Practice: Multiplying Fractions

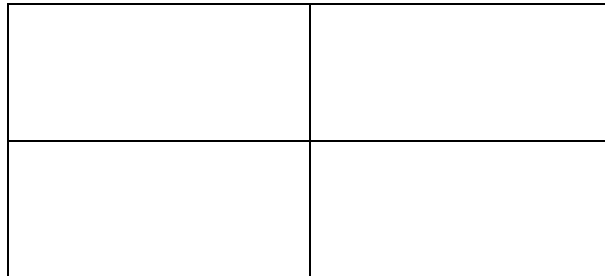
1. Find the product.

a. $6 \times \frac{3}{5}$

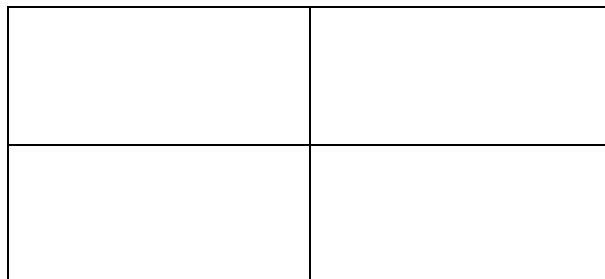
b. $\frac{7}{10} \times \frac{1}{2}$

2. Evaluate each of the expressions using the area model.

a. $2\frac{3}{5} \times 4\frac{1}{3}$



b. $4\frac{7}{8} \times 6\frac{1}{4}$



1.3.4 Collaboration: Multiplying Fractions Greater Than One Using an Algorithm

When finding the product, such as $\frac{7}{10} \times \frac{1}{2}$, the algorithm for multiplying, $\frac{a}{b} \times \frac{c}{d}$, is used.



An **algorithm** is a process or a set of rules to be followed when doing calculations.

1. Work with your partner to complete the equation for the algorithm.

$$\frac{a}{b} \times \frac{c}{d} = \frac{\times}{\times}$$

2. Another way to represent multiplication is to use a dot.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{\cdot}{\cdot}$$

3. Multiplication can also be represented using parentheses.

$$\left(\frac{a}{b}\right)\left(\frac{c}{d}\right) = \frac{(\quad)(\quad)}{(\quad)(\quad)}$$



The algorithm does not account for mixed numbers. To multiply mixed numbers using the algorithm, the mixed number will have to be rewritten as a fraction greater than 1.

4. Work with your partner to complete the statements that explain how to rewrite $6\frac{2}{9}$ as a fraction greater than 1.

Because $6\frac{2}{9} = 6 + \frac{2}{9}$, $6\frac{2}{9}$ can be rewritten by multiplying

$\frac{6}{1}$ by $\frac{9}{9}$ so that 6 is written as an equivalent fraction with a denominator of 9.

This results in $\frac{54}{9} + \frac{2}{9}$ whose sum is $\frac{56}{9}$. Therefore, $\frac{56}{9}$ is $6\frac{2}{9}$ rewritten as a fraction greater than 1.

5. Write $8\frac{5}{7}$ as a fraction greater than 1.

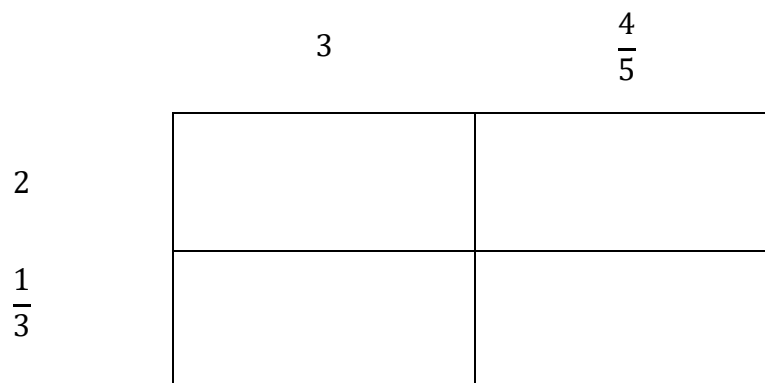
6. Multiply $3\frac{4}{5} \times 2\frac{1}{3}$ by rewriting $3\frac{4}{5}$ and $2\frac{1}{3}$ as fractions greater than 1.

$$3\frac{4}{5} \times 2\frac{1}{3} = \frac{\quad}{5} \times \frac{\quad}{3}$$

7.

a. What is the product of $3\frac{4}{5} \cdot 2\frac{1}{3}$?

b. Verify your answer by completing the area model.



1.3.5 Wrap-Up: Ticket Out the Door

1. Complete the following.

a. What is the product of $6\frac{2}{3} \cdot 7\frac{1}{2}$?

b. Verify your answer by completing the area model.

2. Dan mixes $\frac{1}{4}$ cup of vinegar with a gallon of water to use as a cleaner. How much vinegar does Dan use if he makes 3 batches of the cleaner?